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Investigating antecedents of plagiarism using extended theory of planned behavior

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ABSTRACT

Plagiarism has received considerable attention over the past two decades. Exploring the predictors of this type of academic misconduct can support stakeholders when confronting and managing the incidents of plagiarism. Using an extension of the Theory of Planned Behavior (TPB) framework, this study aims to understand the antecedents of plagiarism. In the study, the influence of new variables such as moral obligation and past behavior were tested along with more familiar TPB constructs. Moreover, the variable of perceived behavioral control was substituted with the Information and Communication Technologies (ICT) literacy variable, which is measured using Internet, computer and information literacy. Adopting a cross-sectional survey design, the data were collected from 588 university students and analyzed using partial least squares structural equation modeling. The results demonstrated that attitude, information literacy, moral obligation and past behavior were significant predictors of behavior intention to engage in plagiarism, whilst subjective norms, Internet literacy and computer literacy were not. The study concluded that the most appropriate way to combat plagiarism is through pedagogy. Offering more courses to university students on the subjects of ethics, morality and literacy is therefore highly recommended.

1. Introduction

Digital technologies offer many possibilities for teaching and learning. For example, students prefer the Internet (e.g., electronic databases or Google) as a major resource to support their academic assignments instead of physical sources found in libraries (Davies & Howard, 2016; Sutherland-Smith, 2008). Accordingly, some have claimed that the Internet has exacerbated the issue of plagiarism (Ma, Wan, & Lu, 2008; Underwood & Szabo, 2003), while others argue that it is not known whether the Internet has increased plagiarism, since no empirical data has corroborated this claim (Davies & Howard, 2016). Nevertheless, what is well known is that plagiarism has an association with the Internet and that students who already plagiarized now use the Internet as a medium to satisfy their plagiarism needs (Camara, Eng-Ziskin, Wimberley, Dabbour, & Lee, 2017; Park, 2003; Sutherland-Smith, 2008).

Plagiarism is a serious concern for universities. Although administrators and faculties take certain precautions, the issue is still prevalent, causing professionals and scholars to look for new ways to combat this unwanted behavior (Cronan, Mullins, & Douglas, 2018). Some innovative technological solutions (e.g., Turnitin, iThenticate) aim to bolster academic integrity by checking the content of documents for their originality, and comparing the similarities of the text to that of documents hosted on other websites and databases as a means to detect plagiarism (Balbay & Kilis, 2019; Bruton & Childers, 2016). Recently plagiarism rates seem to have

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decreased, thanks in part to the potential of detection software (Curtis & Vardanega, 2016). In essence, avoiding and reducing plagiarism requires a holistic institutional approach that includes all stakeholders (students, instructors, institutions, academic managers, and other agencies) through with a shared responsibility approach rather than detection and punishment (Macdonald & Carroll, 2006). Clarifying factors that contribute to the elimination of plagiarism may be helpful for stakeholders to better combat problems arising from this type of academic misconduct. To this end, the current study aims to identify factors that affect university students' intention to plagiarize. In order to model plagiarism intention, the current study refers to the extended Theory of Planned Behavior (TPB) (Ajzen, 1991) by incorporating new variables of moral obligation and past behavior and using ICT literacy as a proxy for perceived behavioral control.

2. Literature review

This first subsection presents what is known about plagiarism, and then goes on to explain the extended theory of planned behavior upon which this study is grounded, and the variables included in the study.

2.1. Plagiarism, prevalence and the reasons behind it

Plagiarism refers to "copying (or using) of others' work that (accidently or otherwise) deceives a third party about the authorship (or ownership) of the work" (Yeo, 2007, p. 201). Students plagiarize intentionally or unintentionally. While intentional plagiarism is premediated deliberate action, unintentional plagiarism occurs accidently or mistakenly due to the students' lack of knowledge about proper referencing (Park, 2003). Although the concept of plagiarism is not new, the way that students plagiarize has changed. The Internet has changed the notion of authorship, which previously was limited to stable printed documents, as it is difficult to embrace conventional authorship notions for the Internet, where information changes, items are relocated from their original publication resource, as well as things being shared with literally anyone easily and ubiquitously (Sutherland-Smith, 2008). Technology equipped with Internet access has made it "easier to steal and cheat, or to otherwise deceive and defraud others" (Stephens, Young, & Calabrese, 2007, p. 234). Students can easily download or copy-paste materials from the Internet and use them within assignments purported as being their own, and without providing the necessary acknowledgement to the author (Akbulut, Şendağ, et al., 2008; Balbay & Kilis, 2019; Jereb et al., 2018; Ma et al., 2008).

Studies have investigated the prevalence of plagiarism, although it is considered difficult to reveal a consistent pattern due to a variety of reasons (Park, 2003). For instance, Selwyn (2008) found that nearly 61.9% of students engaged in some form of plagiarism. Students were found to use digital more often than conventional sources in order to paraphrase or copy the work of others without granting academic acknowledgement (Stephens et al., 2007). Eret and Ok (2014) investigated Turkish teacher candidates' disposition to commit plagiarism through the Internet and concluded that 80.1% of the students who plagiarized used the Internet, albeit at different levels.

A variety of factors can influence plagiarism. Research has explored the influence of gender, age, year of study, and department as individual factors (Bokosmaty, Ehrich, Eady, & Bell, 2019; Chen & Chou, 2017; Eret & Ok, 2014; Jereb, Urh, Jerebic, & Šprajc, 2018; McCabe & Trevino, 1997), in addition to peer behaviors, policy awareness, severity of penalties, and existence of an honor code as situational or contextual factors (McCabe & Trevino, 1997; McCabe, Trevino, & Butterfield, 2002). Attitudes toward class, desire for better grades, peer pressure, time pressure, and workload are some other antecedents of plagiarism addressed by the literature (Akbulut, Şendağ, et al., 2008; Akbulut, Uysal, Odabasi, & Kuzu, 2008; Park, 2003; Stephens et al., 2007). With regards to student cheating, individual differences explained little variance, whereas students' decisions were mostly determined by contextual factors (McCabe & Trevino, 1997). Furthermore, numerous reasons lead to plagiarism, as it is a multidimensional complex issue (Ehrich, Howard, Mu, & Bokosmaty, 2016). Therefore, this study employed extended TPB (Ajzen, 1991) in order to model plagiarism intention.

2.2. Theory and predictions

Ajzen (1991) proposed TPB to model the determinants of a person's behavior. TPB is an extension of Theory of Reasoned Action (TRA), which was initially formed by Ajzen and Fishbein (1980). Ajzen (1991) incorporated perceived behavioral control to TRA to predict conditions when individuals have incomplete volitional control over a behavior, and reformed TRA as TPB.

According to the TPB, a person's actual performance of a behavior is determined by their intention to perform that behavior. Intentions are affected by the combination of the *behavioral*, *normative* and *control beliefs*. Behavioral beliefs refer to consequences and other behavioral attributes, normative beliefs include normative expectations of significant others, whilst control beliefs refer to the existence of factors that facilitate or hinder the performance of a behavior. Behavioral beliefs lead to positive and negative *attitudes toward the behavior*. Normative beliefs lead to *subjective norms*, which could be also conceptualized as perceived social pressure. Control beliefs lead to *perceived behavioral control*, which is the perceived ease or difficulty of performing a behavior. In general, TPB suggests that more favorable attitudes and subjective norms with regard to the behavior, and greater extent of perceived behavioral control lead to a greater intention to perform a behavior (Ajzen, 1991).

The TPB has been extended based on the idea that the addition of variables (e.g., moral obligation and past behaviors) may enhance the predictive accuracy of the model (Ajzen, 1991; Beck & Ajzen, 1991; Cronan & Al-Rafee, 2008; Cronan et al., 2018). Considering the relative importance of such variables on studies dealing with ethical decision-making, the current study will adopt an extended TPB. More specifically, the current study will also evaluate the predictive power of moral obligation and past behaviors as

well as individuals' behavioral (attitude toward the behavior), normative (subjective norms) and control (perceived behavioral control) beliefs. The following subsection refers to the components of the extended TPB in detail.

2.2.1. Attitude

Attitude toward a behavior refers to "the degree to which a person has a favorable or unfavorable evaluation or appraisal of the behavior in question" (Ajzen, 1991, p. 188). Attitude is a crucial factor in ethical decision-making studies. Individuals decide what is ethical or not based on their beliefs and attitudes (Namlu & Odabasi, 2007). Many studies have indicated that attitudes affect plagiarism intention. For instance, among strength of self-control, attitude, and perceived opportunity attitude explained about 40% of the total variance in predicting academic dishonesty (Bolin, 2004). A meta-analysis study showed that students having positive attitudes towards plagiarism were more likely to plagiarize (Whitley, 1998). Recently, it was found that negative attitudes have significantly negatively predicted plagiarism intention, whilst positive attitudes significantly and positively predicted it (Camara et al., 2017). In some studies, students were reported to commit plagiarism, although they believed such practices to be unethical (Hosny & Fatima, 2014). Another recent study presented a critical review of studies on attitudes toward plagiarism. It was concluded that many studies lacked in-depth analysis between attitude and various forms of plagiarism, and few studies were conducted in Asian, Asia-pacific or Middle Eastern contexts. Finally, the study suggested that advanced analysis methods (covariance-based – structural equation modeling (CB-SEM), partial least squares – structural equation modeling (PLS-SEM)) might be used to reveal how plagiarism attitude could be explained in terms of different types of plagiarism (Husain, Al-Shaibini, & Mahfoodh, 2017). Another systematic review study explored the psychological causes of plagiarism and reported that there were a variety of reasons shaping students' attitudes toward plagiarism, resulting in studies not having reached a consensus about the association between them (Moss, White, & Lee, 2018).

2.2.2. Subjective norms

Subjective norms refer to a person's "perceived social pressure to perform or not to perform the behavior" (Ajzen, 1991, p. 188). However, also known as injunctive norms, this definition is regarded as the traditional conceptualization of the social norms that undermines descriptive norms. Whereas descriptive norms relate to the perception of how other people in fact behave (i.e., what is done), injunctive norms pertain to the perception of what is approved or disapproved of by others (i.e., what ought to be done) (Cialdini, Reno, & Kallgren, 1990). Although both descriptive and injunctive norms have separate effects on behavior, research has mostly focused on the injunctive norms (Rajah-Kanagasabai & Roberts, 2015). Research demonstrated that peer-related variables have considerable influence on behaving ethically. For example, a large sized study conducted in nine universities revealed that "the most powerful influential factors were peer-related contextual factors" (McCabe & Trevino, 1997, p. 391). More specifically, academic dishonesty tended to occur less when students perceived their peers disapproved of such misconduct. A meta-analysis study revealed a large effect ($d = 0.929$) for the social norms variable, suggesting that individuals who perceived that their social environment condoned cheating were more likely to cheat (Whitley, 1998). These results suggest that a cheating culture of any social setting may affect a student's cheating behavior (Engler, Landau, & Epstein, 2008).

2.2.3. Perceived behavioral control

Perceived Behavioral Control (PBC) refers to "the perceived ease or difficulty of performing the behavior" (Ajzen, 1991, p. 188). In general, perceiving to have more resources and opportunities and to anticipate less obstacles or impediments are associated with having more PBC (Ajzen, 1991). PBC was found to be a significant and positive determinant of cheating (Camara et al., 2017; Cronan et al., 2018; Mayhew, Hubbard, Finelli, Harding, & Carpenter, 2009; Stone, Jawahar, & Kisamore, 2010), while some studies found no significant effect (Harding, Mayhew, Finelli, & Carpenter, 2007; Passow, Mayhew, Finelli, Harding, & Carpenter, 2006).

As both PBC and self-efficacy include the efficacy dimension, they are regarded as similar constructs (Ajzen, 2002a). Considering this rationale, ICT literacy self-efficacy could be considered a useful determinant that could be used as a proxy for PBC for different reasons. First, ICT literacy includes computer literacy and Internet literacy dimensions, which denote perceived capability or skills to operate different computer and Internet technologies (Lau & Yuen, 2014a). Second, ICT literacy is associated with self-efficacy and thus can be used in place of PBC in TPB models (Chan, 2015). Third, ICT-mediated plagiarism should be clarified, as younger generations (i.e., digital natives) have an inherent ability to use ICT fluently, which may provoke them to engage in plagiarism (Jereb et al., 2018). Finally, students with more experience in using computers reported greater tendencies to plagiarize (Eret & Ok, 2014). Based on the aforementioned literature, the current study uses ICT literacy self-efficacy instead of PBC, which has three dimensions; Information Literacy (INFL), Internet Literacy (INTL), and Computer Literacy (COMPL) (Lau & Yuen, 2014a). While INTL and COMPL are related to the performing of technical skills such as using the Internet and computers effectively, INFL denotes a higher-order skill related to research and communication through technological means (Katz, 2005).

Although INTL and COMPL provide certain advantages in many aspects, it could also be misused. More specifically, it is probable that students with high proficiency in INTL and COMPL could easily access information from different digital sources and may manipulate the contents without the provision of acknowledgement to the owner(s). On the other hand, INFL is different from INTL and COMPL in that "to be information literate, a person must be able to recognize when information is needed and have the ability to locate, evaluate and use effectively the needed information" (Katz, 2005, p. 3). Therefore, INFL may function differently in predicting plagiarism. Plagiarism might occur in circumstances when students lack sufficient literacy skills including poor research, reading and writing (Davies & Howard, 2016; Howard & Davies, 2009; Park, 2003). For example, Rodrigue, Serviss, and Howard (2007) observed students' assignments written in the research writing course and found that all of the papers contained abuse of the sources, and deficient summary strategies. In a similar vein, Norton, Tilley, Newstead, and Franklyn-Stokes (2001) found that students who did

not use evidence or logic were found more likely to engage in cheating.

2.2.4. Moral obligation and past behavior

Although TPB is a useful model to forecast behavioral intention, the addition of other antecedents was suggested in order to increase the predictive power of the model. One of the newly added antecedents is moral obligation, which denotes the use of "personal feelings of moral obligation or responsibility to perform, or refuse to perform, a certain behavior" (Beck & Ajzen, 1991, p. 289). Moral obligation implies one's feeling of guilt or obligation to moral principles related to performing or not performing a behavior. It could be considered as a useful determinant, as academic dishonesty behaviors such as homework cheating and plagiarism contain moral aspects that lead individuals to feel guilt or to consider their obligations to moral values (Cronan et al., 2018).

Moral obligation is deemed to be crucial, since the act of plagiarism is seen as a direct relation. Research has indicated that the inclusion of moral obligation in the TPB increases the prediction of ethical intention (Beck & Ajzen, 1991; Cronan & Al-Rafee, 2008; Cronan et al., 2018). A meta-analysis study showed moral obligation has a medium effect ($d = -763$) on plagiarism intention (Whitley, 1998). In another study, Armitage and Conner (1999) demonstrated that the relationship between moral norms and other components of the TPB were significant. Passow et al. (2006) indicated that students having the belief that cheating is wrong were also inclined to cheat less, independent of the context. Additionally, a recent study indicated that students who considered the performance of a certain action to be morally wrong also reported less involvement in undertaking that action (Stephens, 2018).

Past behavior is another factor that has attracted researchers' attention in the prediction of intention. Many behaviors are designated as past behaviors, rather than the cognitions illustrated in TPB (Sutton, 1994, as cited in Conner & Armitage, 1998). Besides, the inclusion of past behavior has been shown to have a significant influence on behavior intention (Bagozzi & Kimmel, 1995; Bentler & Speckart, 1981). However, in some studies, the inclusion of past behavior has seemed to attenuate relationships among the TPB components (Hagger, Chatzisarantis, & Biddle, 2002). Additionally, Ajzen (2002b) argued that past behavior may affect subsequent behaviors independently of intention, and that this could be explained by the habitual phenomena. Thus, "past behavior frequency adds little to our understanding of a behavior's determinants" (Ajzen, 2002b, p. 120). Then again, it was argued that the distinction between past behavior and habit is unclear (Conner & Armitage, 1998). Also, past behavior was shown to have its own unique effect on intention (Towler & Shepherd, 1991). Having cheated in the past had one of the strongest correlations with cheating behavior or subsequent cheating (Harding et al., 2007; Passow et al., 2006; Whitley, 1998). More recently, past behavior was found as the second strongest predictor after attitude in predicting plagiarism violation intention (Cronan et al., 2018).

Based on the aforementioned literature, we postulate the following research question:

Are students' attitudes, subjective norms, perceived behavioral control (internet literacy, computer literacy, and information literacy), moral obligation, and past behaviors significantly associated with their intention to plagiarize?

In accordance with the research question, the following hypotheses are formulated.

H1. Attitude toward plagiarism is significantly positively associated with intention to plagiarize.

H2. Subjective norms are significantly negatively associated with intention to plagiarize.

H3a. Internet literacy is significantly positively associated with intention to plagiarize.

H3b. Computer literacy is significantly positively associated with intention to plagiarize.

H3c. Information literacy is significantly negatively associated with intention to plagiarize.

H4. Moral obligation is significantly negatively associated with intention to plagiarize.

H5. Past behavior is significantly positively associated with intention to plagiarize.

The following section describes the research methodology including research design, research model with the application of PLS-SEM model and procedure, sample, data collection instruments, and data analysis.

3. Method

The current study adopted quantitative survey research design that aims to analyze a group of people's beliefs, attitudes and abilities. More specifically, a cross-sectional survey design was employed in which the data were obtained at a singular point in time, thus having the advantage of probing the participants' current tendencies (Creswell, 2012; Fraenkel, Wallen, & Hyun, 2012).

3.1. Research model

Following the formulated hypotheses, it was supposed that attitude, subjective norms, ICT literacy, moral obligation, and past behaviors were antecedents of plagiarism intention. ICT literacy dimension was substituted for perceived behavioral control in order to probe the influence of distinct literacy dimensions of Internet Literacy (INTL), Computer Literacy (COMPL), and Information Literacy (INFL) on plagiarism intention. New antecedents (i.e., moral obligation and past behavior) were included, which have been discussed widely in the literature (Cronan & Al-Rafee, 2008; Cronan et al., 2018; Harding et al., 2007; Passow et al., 2006). The proposed model is illustrated in Fig. 1.

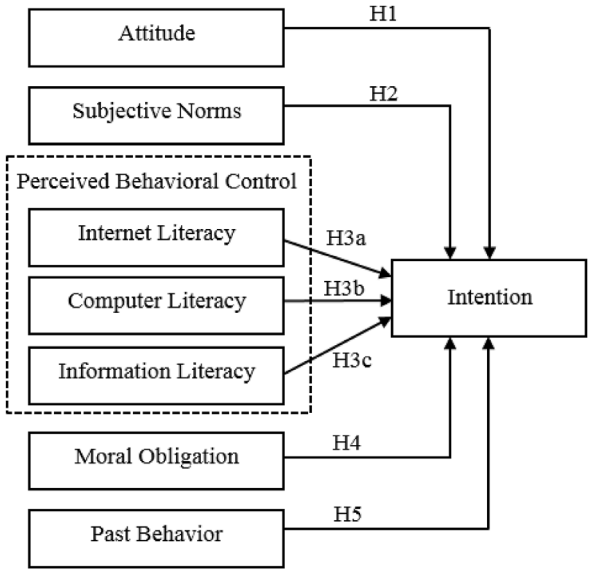


Fig. 1. Research model.

3.2. Participants of the study

The participants were university students from various disciplines at the Faculty of Education of a state university located in the Aegean region of Turkey. The study adopted convenience sampling, which is a non-probability sampling method based on willingness, convenience and availability of the participants (Creswell, 2012). The target population consisted of 1,956 students. Initially, data were collected from 642 participants out of whole population. After excluding the cases which had been partially completed or where the same rating had been applied to all of the survey items, the final sample was reduced to 588 participants.

In terms of the demographic information of the participants, the majority were female (n = 440; 75%), and the minority were male (n = 148; 25%). Their mean age was 20.77 years (SD = 1.47). As for their class distribution, most of them were juniors (n = 224; 38%), followed by seniors (n = 201; 34%), and then sophomores (n = 163; 28%). No data were collected from freshman students because, at the time of the data collection, the freshmen students had only been studying at the university for a couple of weeks.

3.3. Data collection procedure and instruments

The data were collected during the fall semester of the 2018–2019 academic year. Considering the measurement tool contains items regarding the sensitive issue of "plagiarism," prior to applying the survey, the participants were assured that their anonymity and confidentiality would be protected.

Academic Integrity (AI) violation survey (Cronan et al., 2018) and ICT literacy scale (Lau & Yuen, 2014a) were employed in the current study as data gathering tools. Cronan et al. (2018) incorporated the results of various research studies in order to generate the survey's items. More specifically, intention (Madden, Ellen, & Ajzen, 1992), attitude (Bodur, Brinberg, & Coupey, 2000; Madden et al., 1992), subjective norms (Ajzen, 1991), and moral obligation (Beck & Ajzen, 1991) were derived from other research, whereas past behavior was developed by Cronan et al. (2018) themselves. The current study applied the (AI) violation survey and adapted it to fit the current study's context. In order to make the plagiarism items compatible with the scope of the current study (plagiarism), other researchers' works were also incorporated (Akbulut, Şendağ, et al., 2008; Stephens et al., 2007). In addition, the ICT literacy scale was used as a proxy for the perceived behavioral control dimension of Theory of Planned Behavior. As the participants were all Turkish speakers, the survey items were translated and adapted to the Turkish language context. In the translation process, the items were first translated into Turkish by four scholars, two of whom were experts in Educational Sciences and two in Educational Technology. After that, the items were reverse-translated from Turkish into English by a scholar from the Department of English Language. Then, similarity of the original and adapted items were checked and final form items were agreed. As a final step, a Turkish language expert revised the items in terms of wording, grammar, and clarity. The data collection tool used in the study is included in the Appendix of this study.

Intention was measured by three items, rated on a 7-point, Likert-type scale (definitely [1]/definitely not [7]). High scores refer to a greater inclination to engage in plagiarism in the near future.

Attitude was measured by four bipolar semantic differential items. Students rated their attitudes by answering semantic differential items on a 7-point, Likert-type scale. The four items were favorable (1)/unfavorable (7), harmful (1)/beneficial (7), foolish (1)/wise (7), and good (1)/bad (7). High scores indicate more favorable attitudes towards plagiarism.

Subjective norms were assessed by three items. Students rated their beliefs about how “significant others” affect their plagiarism behavior on a 7-point, Likert-type scale (strongly agree [1]/strongly disagree [7]). A high score indicates higher levels of subjective norms not to commit plagiarism.

Perceived behavioral control was assessed by a three-factor ICT literacy scale, in which the factors were conceptualized as Internet Literacy (INTL), Computer Literacy (COMPL), and Information Literacy (INFL). The scale was originally developed for middle school students by Lau and Yuen (2014a). However, Nasser AL-Nuaimi, Bouazza, Abu-Hilal, and Al-Aufi (2014) adapted the scale by modifying items in a manner that conforms to the undergraduate context, and from which they obtained adequate reliability and validity values. Thus, referring to Nasser AL-Nuaimi et al. (2014), some items were modified in accordance with the context of the current study. INTL includes five items related to technical competencies of using the Internet. Finally, COMPL includes five items that mostly relate to the offline usage of computer technologies and applications. All items are rated on a 5-point, Likert-type scale (strongly disagree [1]/strongly agree [5]). INFL includes seven items that probe information literacy skills related to researching and communicating information (Katz, 2005). High scores obtained from the INTL, COMPL, and INFL scales indicate possessing higher levels of the Internet, computer, and information literacy, respectively.

Moral obligation was assessed by three items, which were about participants' moral obligation to avoid plagiarism. Items were rated on a 7-point, Likert-type scale (strongly agree [1]/strongly disagree [7]). High scores indicate greater moral obligation to abstain from plagiarism.

Past behavior was assessed by two items regarding the degree and frequency of the occurrences of plagiarism. The first item (degree) was rated on a 7-point, Likert-type scale (a lot [1]/very little [7]), whereas the second item (frequency) was rated on a 5-point, Likert-type scale (never to [1]/every time [5]). High scores indicate more occurrences of plagiarism in the past.

3.4. Data analysis

To analyze the data, structural equation modeling was utilized. More specifically, based on the two-step approach (Anderson & Gerbing, 1988), first, the measurement model was tested and then the structural model.

Partial least squares – structural equation modeling (PLS-SEM) with SmartPLS 3.2.7 was used to test both the measurement and structural models. PLS-SEM enables different affordances compared to covariance-based – structural equation modeling (CB-SEM). For example, PLS-SEM is less sensitive to violation of the normality assumption and more suited to circumstances where the primary purpose is to predict the target variable(s). PLS-SEM can be used for complex models including many constructs and factors with less items (e.g., one or two), and can be included in the PLS-SEM models, as it has less restrictive measurement properties than CB-SEM (Hair, Hult, Ringle, & Sarstedt, 2014; Hair, Ringle, & Sarstedt, 2011). In the current study, some variables of the ICT literacy scale were regarded to have inappropriate skewness and kurtosis values (outside of the range of -2 to $+2$), as absolute values greater than 1 indicates a non-normal distribution (Hair et al., 2014). Besides, the current study focused on predicting plagiarism intention rather than theory confirmation and had some constructs with less (e.g., past behavior) items. Given this rationale, the PLS-SEM was deemed to be an appropriate method for the current study.

The measurement model was tested in order to provide evidence of its reliability and validity. Evidence for internal consistency, convergent and discriminant validity were presented. The structural model was explored by evaluating the path coefficients among constructs so as to test the hypothetical relationships.

4. Results

4.1. Testing measurement model

Prior to testing the hypotheses, first the measurement model was analyzed. One item of subjective norms (SN2) and one item of Computer Literacy (COMPL5) were excluded from the analysis as having low factor loadings. All of the remaining items had sufficient factor loadings above the suggested threshold value (0.50) (Fornell & Larcker, 1981) (see Table 1).

Internal consistency reliability was assessed using composite reliability (CR) and Cronbach's Alpha value. However, it was noted that Cronbach's Alpha generally underestimated the exact value of the reliability. Therefore, Dijkstra-Henseler's rho (rho_A), which is the latest and most significant reliability measure of PLS, was also used to achieve a more precise estimation (Dijkstra & Henseler, 2015).

The advised threshold value for CR, Alpha and Dijkstra-Henseler's rho is 0.70 (Hair et al., 2014; Nunnally, 1978). According to Table 1, all constructs had CR values above the threshold value. Besides, all constructs except for SN (0.47) had sufficient Alpha values. Although SN did not have a satisfactory alpha value, Hair et al. (2014) suggested to consider Alpha as a conservative measure of internal consistency reliability. Hence, SN as well as other constructs were deemed to have sufficient internal consistency reliably. The average variance extracted (AVE) values calculated for all constructs were above the suggested threshold value of 0.50, suggesting that convergent validity was met. Discriminant validity was assessed by using Fornell and Larcker’s (1981) criteria, in which squared root of AVE value for a construct is suggested to exceed the correlations between latent constructs to diagnose the potential overlaps (see Table 2).

As can be seen from Table 2, all pairwise correlations between constructs (off diagonal values) were lower than the squared root of AVE values (diagonal values), suggesting that discriminant validity had been established. However, this criterion does not calculate discriminate validity accurately in certain circumstances (Henseler, Ringle, & Sarstedt, 2015). Therefore, Hetrotrait – Monotrait (HTMT) ratio of correlations method was also applied (Henseler et al., 2015). Based on this method, HTMT values should

Table 1
Analysis of measurement model.

Construct	Item	Loading	AVE	CR	Alpha	Rho_A
ATT			.68	.90	.85	.89
	ATT1	.86				
	ATT2	.80				
	ATT3	.78				
	ATT4	.87				
COMPL			.69	.90	.85	.90
	COMPL1	.88				
	COMPL2	.85				
	COMPL3	.85				
	COMPL4	.74				
INFL			.63	.92	.90	.91
	INFL1	.73				
	INFL2	.82				
	INFL3	.84				
	INFL4	.81				
	INFL5	.82				
	INFL6	.83				
	INFL7	.71				
INT			.86	.95	.92	.92
	INT1	.92				
	INT2	.96				
	INT3	.91				
INTL			.58	.87	.82	.82
	INTL1	.79				
	INTL2	.80				
	INTL3	.73				
	INTL4	.65				
	INTL5	.82				
MO			.66	.85	.74	.75
	MO1	.80				
	MO2	.85				
	MO3	.78				
PB			.81	.89	.77	.78
	PB1	.92				
	PB2	.88				
SN			.61	.75	.47	.89
	SN1	.55				
	SN3	.96				

(ATT: Attitude, SN: Subjective norms, COMPL: Computer literacy, INFL: Information literacy, INT: Intention, INTL: Internet literacy, MO: Moral obligation, PB: Past behavior).

Table 2
Results of discriminant validity (based on Fornell and Larcker's (1981) criteria).

	ATT	SN	COMPL	INFL	INT	INTL	MO	PB
ATT	**.83**							
SN	-.30	**.78**						
COMPL	-.12	.02	**.83**					
INFL	-.21	.14	.54	**.80**				
INT	.43	-.35	-.32	-.32	**.93**			
INTL	-.18	.05	.64	.73	-.24	**.76**		
MO	-.47	.54	.08	.23	-.56	.13	**.81**	
PB	.34	-.31	-.15	-.19	.61	-.18	-.47	**.90**

(ATT: Attitude, SN: Subjective norms, COMPL: Computer literacy, INFL: Information literacy, INT: Intention, INTL: Internet literacy, MO: Moral obligation, PB: Past behavior).

be lower than the most conservative value of 0.85, especially in circumstances where the constructs theoretically differ. However, the liberate value of 0.90 is also acceptable, especially if the latent variables measure theoretically similar constructs (Henseler et al., 2015). In the current study, all HTMT values were below 0.85, except the calculation for INTL and INFL which were 0.86. However, as these latent variables measure similar constructs and being sub-factors of ICT literacy, a liberate value of 0.90 was applied for them.

Additionally, as suggested by Henseler et al. (2015), the full bootstrapping procedure was applied to test null hypothesis (H_0: HTMT $\geq$ 1) against the alternative hypothesis (H_1: HTMT $<$ 1). Confidence interval values involving a value of 1 indicates that

Table 3
Results of structural model.

Hypothesis	Relationship	β	t	Decision	2.50%	97.50%	VIF	R^2	Q^2	f^2
								.51	.42	
H1	ATT→INT	.13	2.93*	Supported	.05	.22	1.33			.026
H2	SN→INT	-.01	.25	Rejected	-.08	.06	1.44			.000
H3a	INTL→INT	-.02	.38	Rejected	-.12	.08	2.70			.000
H3b	COMPL→INT	.03	.78	Rejected	-.05	.11	1.74			.001
H3c	INFL→INT	-.15	3.31*	Supported	-.24	-.06	2.31			.021
H4	MO→INT	-.27	5.98**	Supported	-.36	-.18	1.86			.081
H5	PB→INT	.41	11.96**	Supported	.35	.48	1.35			.260

Notes: *$p < .01$ = Significant; **$p < .001$ = Significant.
(ATT: Attitude, SN: Subjective norms, COMPL: Computer literacy, INFL: Information literacy, INT: Intention, INTL: Internet literacy, MO: Moral obligation, PB: Past behavior).

discriminant validity is not established. In the current study, no values were greater than 1 in either the lower or upper confidence intervals. Thus, the discriminant validity was established. In conclusion, the measurement model generated sufficient levels of reliability, convergent and discriminant validity suggesting that structural model was fit to be tested.

4.2. Testing structural model

In order to test the structural model, PLS-SEM with SmartPLS 3.2.7 was used. To test the statistical significance of estimated path coefficients (hypothesis), Hair et al. (2014) suggested performing bootstrapping resampling routine with 5,000 subsamples. Mainly, R^2 value, beta coefficient (β), significance of the path coefficients and corresponding t values should be reported. However, reporting effect sizes is as important as reporting p values, because "while a p value can inform the reader whether an effect exists, the p value will not reveal the size of the effect" (Sullivan & Feinn, 2012, p. 279). Therefore, Hair et al. (2014) also suggested to report Q^2 and f^2 which denote predictive relevance and effect size, respectively. Table 3 presents the results of the full structural model.

According to Table 3, overall, the R^2 was calculated as 0.51, suggesting that the model explained 51% of variance in plagiarism intention. R^2 values of 0.67, 0.33, and 0.19 indicate substantial, moderate, and weak model, respectively (Chin, 1998). Based on Chin's suggestion, the strength of the model in predicting intention was moderate. After calculating the R^2 value, significance of the path coefficient was tested.

H1 postulates that attitudes toward plagiarism are significantly associated with behavior intention to plagiarize. The results showed that attitude had a significantly positive influence on intention to plagiarize ($\beta = 0.13$, $t = 2.93$, $p < .01$). It could be concluded that students who have more favorable attitudes toward plagiarism tend to exhibit more intention to plagiarize; therefore, H1 was accepted.

H2 posits that subjective norms negatively predict intention. The results showed that subjective norms did not significantly predict intention, although the direction of the relationship was negative; hence, H2 was rejected.

In H3, the influence of ICT literacy as a proxy for perceived behavioral control was tested. For each dimension of ICT literacy, three hypotheses were formulated. H3a assumes that Internet literacy (INTL) is significantly positively associated with intention. The results indicated that INTL did not significantly relate to intention; therefore, H3a was rejected. H3b argues that computer literacy (COMPL) is significantly positively associated with intention to commit plagiarism. The findings demonstrated that COMPL did not significantly predict intention; therefore, H3b was rejected. H3c postulates that INFL is significantly negatively associated with intention. Results showed that INFL had a significant negative influence on intention to plagiarize ($\beta = -0.15$, $t = 3.31$, $p < .01$); hence, H3c was accepted.

In H4, moral obligation is predicted to be significantly negatively related to intention. Results indicated that moral obligation to avoid plagiarism significantly negatively predicted intention ($\beta = -0.27$, $t = 5.98$, $p < .001$); therefore, H4 was accepted.

In H5, past behavior is predicted to be significantly related to intention. Results demonstrated that past behavior had a significant positive influence on behavior intention to plagiarize ($\beta = 0.41$, $t = 11.96$, $p < .001$); therefore, H5 was accepted.

The results of the whole model are illustrated in Fig. 2.

This study reports the practical significance (effect size) as well as the statistical significance (p values) of the results. For this purposes, Q^2 (predictive relevance) and f^2 (effect size) were also reported. To evaluate Q^2, blindfolding procedure was employed. Q^2 values greater than zero suggest that the model has predictive relevance (Hair et al., 2014). As Q^2 was calculated as 0.42 in the current study, predictive relevance was deemed to have been established.

To investigate the relative importance of each construct, f^2 values were examined. f^2 values of 0.02, 0.15, and 0.35 f^2 values represent small, medium, and large effects, respectively (Cohen, 1988), of the exogenous latent variables. According to Table 3, SN, INTL, and COMPL did not have any effect in explaining variance on the intention to commit plagiarism. On the other hand, PB was found to be the most influential factor for intention ($f^2 = 0.260$), indicating a medium to strong effect. MO was the second most influential factor ($f^2 = 0.081$), indicating a small to medium effect, whilst ATT ($f^2 = 0.026$) and INFL ($f^2 = 0.021$) had almost equal values, indicating a small effect.

Finally, model fit was assessed by SRMR (Standardized Root Mean Square Residual), which is currently the only approximate

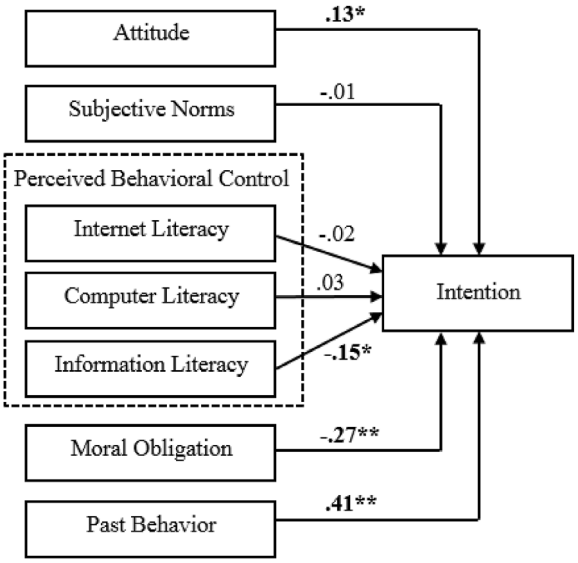


Fig. 2. Results of the structural model.

model fit indicator for PLS (Henseler, Hubona, & Ray, 2016). SRMR values less than 0.08 are acceptable (Hu & Bentler, 1999). In the current study, the SRMR value was 0.056, which indicated a good fit.

5. Discussion

This study aimed to identify factors affecting students' intention to plagiarize. The extended TPB was used as a model to explain behavioral intention to engage in plagiarism using digital sources. To summarize, the results showed that attitude, information literacy, moral obligation, and past behavior were all significant predictors of intention to plagiarize, whilst subjective norms, Internet literacy, and computer literacy were not. As for the strength of the effect, past behavior and moral obligation were found to have the strongest effects on intention.

Regarding to the distinct effects of TPB variables, attitude significantly predicted intention to commit plagiarism. Having a more positive attitude leads to increased inclination to plagiarize, which concurs with the literature on plagiarism intention (Camara et al., 2017; Cronan et al., 2018; Jordan, 2001), and also with other studies that focused on other unethical behavior intention such as digital piracy (Akbulut, 2014; Cronan & Al-Rafee, 2008). Students may have believed that plagiarism was not a serious offense, considering information published on the Internet to be in the public domain and therefore has no need for citation or credit (Jung, 2009; Lau & Yuen, 2014b).

Subjective norms were not found to be a significant predictor of intention, though they were negatively associated with it. In other words, students' ethical decision-making regarding plagiarism was not linked with the beliefs of their significant others. Even though this result differs from some earlier studies (Eret & Ok, 2014; McCabe & Trevino, 1997; Whitley, 1998), it is consistent with others (Beck & Ajzen, 1991; Cronan & Al-Rafee, 2008; Cronan et al., 2018).

The non-significant influence of subjective norms may be due to, “unlike the health, safety, and conservationism-related behaviors that are positive and generally consistent with social norms, cheating and lying run counter to norms and rules” (Stone, Jawahar, & Kisamore, 2009, p. 223). Another reason behind this non-significant association maybe that descriptive norms were not measured in the current study. Therefore, subjective norms as measured by injunctive norms might have failed to make a significant contribution to the model.

It may also be due to the attenuation effect, as inclusion of additional variables might attenuate the relationship between TPB constructs (Beck & Ajzen, 1991; Hagger et al., 2002; Stone et al., 2010). The model in the current study explained increased variance (51%) in the scenario where moral obligation and past behavior were included, than the scenario where they were excluded (28%). This result also corroborates previous literature, which found that adding moral obligation and past behavior additional variables increased the strength of predictability of intention, but attenuated the relationship between subjective norm and intention (Beck & Ajzen, 1991; Cronan et al., 2018; Kurland, 1995).

A unique contribution of the current study was that perceived behavioral control was assessed using three literacy measures. The results demonstrated that Internet literacy and computer literacy were not significant predictors. Even though this result differs from some earlier studies (Eret & Ok, 2014; Selwyn, 2008; Trushell, Byrne, & Hassan, 2013), which found technological capability to be related to higher inclination of plagiarism, it is consistent with other studies which found no significant association (Byrne & Trushell, 2013; Chan, 2015; Kim, Kim, & Lee, 2014; Stephens et al., 2007). Based on these results, one can argue that students' intention

towards plagiarism may need only to be based on copy and paste, which is a very simple skill that almost anyone can perform (Byrne & Trushell, 2013; Mavrinac, Brumini, Bilić-Zulle, & Petrovečki, 2010). To put it in a different way, students may not feel the need for complex technical skills related to using the Internet and computers in order to plagiarize from digital sources. Another reason might be that tech-savvy students might believe that plagiarism is not egregious when compared to hacking or digital piracy (Chan, 2015).

On the other hand, when it comes to the influence of information literacy on intention, the findings highlighted that it significantly negatively predicted intention. This lends support to some earlier studies, which reported that low information literacy skills might lead to plagiarism (Davies & Howard, 2016; Howard & Davies, 2009; Norton et al., 2001). This might be because students with low information literacy consume the information passively without reading it; being more interested in saving time by obtaining information from the Internet without paying attention to ethical considerations. Additionally, when critical thinking is low, information seeking is poor which leads to increased inclination to plagiarize (Weiler, 2005).

As for moral obligation, it was found to be significantly related to intention to plagiarize, with a small to medium effect size. Students who consider plagiarism to be morally wrong tend to report less intention to commit such a behavior. This lends supports to previous findings, which found moral values to be negatively related to cheating (Cronan et al., 2018; Harding et al., 2007; Passow et al., 2006; Stephens, 2018; Stephens et al., 2007). The current study's result further supports the idea that moral obligation could be integrated into the TPB model, especially for the research context, where dishonest behaviors are considered to involve moral aspects (Beck & Ajzen, 1991; Cronan et al., 2018; Mayhew et al., 2009). Despite an argument that inclusion of moral obligation in the TPB model may be of little practical value (Beck & Ajzen, 1991), the current study found it to be the second most important factor in terms of the strength of the effect. This is not surprising considering that moral obligation is a critical determinant of behavior intention for most unethical dishonest behaviors, regardless of the context (Harding et al., 2007; Mayhew et al., 2009; Passow et al., 2006).

As for past behavior, it was also found to significantly positively predict behavior intention, with a medium to strong effect. Among all other variables, it was the strongest predictor of intention, which is in solid agreement with previous research findings (Cronan et al., 2018; Harding et al., 2007; Passow et al., 2006; Whitley, 1998), and is also consistent with the recognition that it is the best predictor of future behavior (Owens & Schoenfeldt, 1979). One critical result was that considering previous success at cheating might lead to more favorable attitudes toward cheating; resulting in a misperceived peer approval of cheating, and thereby curtail feelings of moral obligation to abstain from cheating (Whitley, 1998). To summarize, as previously mentioned, many current behaviors are determined by past behaviors, rather than the cognitions shown in the TPB (Sutton, 1994, as cited in Conner & Armitage, 1998).

5.1. Implications for practice

A few important implications could be suggested based on the results. One way to diminish plagiarism would be to work towards the changing of attitudes compared to personality traits (Bolin, 2004). Attitudes could be alterable by means of various educational interventions (Ames & Eskridge, 1992; Kilis & Uzun, 2018). Students' attitudes could be changed by informing them about what specifically constitutes as plagiarism, and also the consequences of doing so. Additionally, campaigns, conferences and seminars could be arranged in order to increase the awareness of students. A master-apprenticeship system or program could be invoked so as to teach students how to reap appropriate benefit from different resource types in accordance with academic ethical standards of best practice. In order to help students become much more attentive in dealing with their assignments, academic integrity should be promoted as an integral part of all teaching and learning activities.

Academic integrity is also crucial for teacher education programs, considering that preservice teacher graduates might well practice what they observe and learn from their instructors. Teacher educators are therefore principally responsible for the teaching of moral virtues to their students, who are, after all, candidate teachers themselves. At this point, faculty instructors need to act as role models for student teachers. For example, they should demonstrate how they cite others' works they use as course materials (Kilis & Uzun, 2018; Sutherland-Smith, 2008).

In universities, a special focus should be devoted to the relationship between information literacy and plagiarism. Students can be instructed in the critical skill of assessing what information and sources they need, and how to interpret them based on the authority, bias, and timeliness of those materials (Katz, 2005; Weiler, 2005).

Institutions should place emphasis on moral education in order to teach moral virtues such as responsibility, honesty, trust and fairness, and that this should start as early as elementary education (Kilis & Uzun, 2018; Stephens et al., 2007). At the university level, establishing a campus moral code and increasing student awareness could prove useful in founding a moral climate, as educational institutions having honor codes confront fewer incidents of plagiarism (McCabe et al., 2002). It is also important to highlight that rather than preventing regulations, raising awareness and promotion-focused strategies could work better in dealing with this misconduct (Akbulut & Dönmez, 2018; Moon, Kim, Feeley, & Shin, 2015). Moreover, the earlier educators intervene in the commitment of plagiarism, the less occurrences of such behaviors may be observed in their later career, which places emphasis on early diagnosis. As Stone et al. (2010) argued, “Students must learn about the consequences of unethical acts in the formative years of their professional development when consequences to the individual actor are significant but consequences to others are low” (p. 236).

5.2. Limitations and future directions

The current study clearly has some limitations. Potentially, the most evident limitation relates to social desirability bias. Although students were assured of their anonymity and data privacy in terms of taking part in the current study, their truthfulness in responding to the items was an unknown factor. Additionally, no measures were taken to reveal social desirability bias. Future research

may benefit from such a measure in order to detect whether or not students' responses in any way relate to social desirability bias.

Another limitation is that the current study did not examine the students' study major. Future research may consider this variable in order to reveal how studying within different academic departments may affect students' intention to plagiarize, as students' exposure to types, context, and frequencies of assignments could differ based on their major, and thereby their tendency to plagiarize.

The sample characteristics of the current study also deserve attention. Data were obtained from one specific university; therefore, caution should be advised when generalizing the results. Furthermore, the study group consisted predominantly of female students. Future studies should be replicated with a more gender-balanced sample. Another issue for the current study was the educational stage of the participants. As criticized by Blau and Eshet-Alkalai (2017), most studies on ethical issues have been conducted with university students; therefore, future works with a focus aimed at other students groups are required. Factors affecting students' considerations of ethical issues in their teaching practices, such as how to use others' instructional materials obtained from the Internet, could therefore be examined.

6. Conclusion

Plagiarism, which is a multidimensional complex phenomena, is a serious concern for today's educational institutions. Leveraging the extended theory of planned behavior, the current study investigated the determinants of behavioral intention to engage in plagiarism. The strength of the current study lies in the approach taken – ICT literacy, moral obligation and past behavior were incorporated into the TBP model in order to predict plagiarism.

To summarize the current study, evidence suggested that attitude was a significant factor affecting intention. Students' intention to plagiarize was not seen to be affected by subjective norms, which is an external motivation, but seems to be affected by moral obligation, which is an internal motivation. Plagiarism can be as easy as copying and pasting digital sources, which do not necessitate complex Internet and/or computer skills. Having low information literacy increased the tendency to be involved in this misconduct. As a concluding remark, this study highlights that “Plagiarism is best addressed pedagogically. Only through education can students begin to change and improve how they write” (Davies & Howard, 2016, p. 598). In this study, past behavior was found to be the best predictor of future behavior. The findings point to the necessity of implementing courses that address ethics morality, and literacy, which clearly have the potential to minimize dishonest behaviors such as plagiarism.

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Appendix

Constructs		Items
Attitude (ATT)		In my opinion, copying others' works from a digital source (e.g., the Internet) and using them in my own assignments without acknowledgement is:
	ATT1*	Favorable (1)/Unfavorable (7)
	ATT2	Harmful (1)/Beneficial (7)
	ATT3	Foolish (1)/Wise (7)
	ATT4*	Good (1)/Bad (7)
Computer Literacy (-COMPL)	COMPL1	I am able to set and adjust the page's header - footer and adjust its margins by using a word processing program such as Microsoft Word.
	COMPL2	I am able to plot a graph and chart using a spreadsheet program like Microsoft Excel.
	COMPL3	I am able to include and coordinate animation in a program for presentation, such as Microsoft PowerPoint.
	COMPL4	I am able to edit pictures and videos using the appropriate software.
	COMPL5	I am able to install the computer operating system, software and equipment.
Information Literacy (INFL)	INFL1	I am able to identify appropriately the needed information from question.
	INFL2	I am able to collect/retrieve information in digital environments.
	INFL3	I am able to use ICT to process appropriately the obtained information.
	INFL4	I am able to interpret and represent information, such as using ICT to synthesize, summarize, compare and contrast information from multiple sources.
	INFL5	I am able to use ICT to design or create new information from information already acquired.
	INFL6	I am able to use ICT to convey correct information to appropriate targets.
	INFL7	I am able to judge the degree to which information is practical or satisfies the needs of the task, including determining authority, bias, and timeliness of materials.
Intention (INT)	INT1*	I intend to copy others' works from a digital source (e.g., the Internet) and use them in my own assignments without acknowledgement in the near future.
	INT2*	I will try to copy others' works from a digital source (e.g., the Internet) and use them in my own assignments without acknowledgement in the near future.
	INT3*	I will make an effort to copy others' works from a digital source (e.g., the Internet) and use them in my own assignments without acknowledgement in the near future.

Internet Literacy (INTL)	INTL1	I am able to set a homepage for an Internet browser.
	INTL2	I am able to search for information on the Internet using the advanced search options made available by search engines such as Google, Yahoo etc.
	INTL3	I am able to use email to communicate.
	INTL4	I am able to use instant messaging software (e.g., WhatsApp, Skype, Facebook, Messenger etc.) to chat with friends.
	INTL5	I am able to download files and free software from the Internet.
Moral Obligation (MO)	MO1	I would not feel guilty if I copy others' works from a digital source (e.g., the Internet) and use them in my own assignments without acknowledgement.
	MO2*	Copying others' works from a digital source (e.g., the Internet) and using them in my own assignments without acknowledgement goes against my principles.
	MO3*	It would be morally wrong to copy others' works from a digital source (e.g., the Internet) and use them in my own assignments without acknowledgement.
Past Behavior (PB)	PB1	How much have you copied others' works from a digital source (e.g., the Internet) and used them in your own assignments without acknowledgement in the past few years?
	PB2	In the past 2 years, how frequently did you copy from others' works from a digital source (e.g., the Internet) and use them in your assignments without acknowledgement?
Subjective Norms (SN)	SN1*	Most people who are important to me think I should not copy others' works from a digital source (e.g., the Internet) and use them in my own assignments without acknowledgement.
	SN2*	When considering copying others' works from a digital source (e.g., the Internet) and using them in my own assignments without acknowledgement, I wish to do what people who are important to me want me to.
	SN3*	If I copy others' works from a digital source (e.g., the Internet) and use them in my own assignments without acknowledgement, then most people who are important to me would disapprove.

Notes: * Items were reverse-coded for the analysis. Items in bold were excluded from the analysis due to having low factor loadings.

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Update

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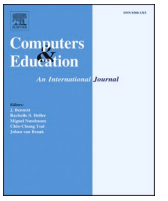
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Corrigendum to "Investigating antecedents of plagiarism using extended theory of planned behavior" [Computers & Education 144 (January 2020) 103700]

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The authors regret for the below corrections:

Correction # 1

2.2. Theory and predictions

2.2.3. Perceived behavioral control

Second, ICT literacy is associated with self-efficacy and thus can be used in place of PBC in TPB models examining ICT ethics-related cognition and behavior (Chan, 2015; Nasser Al-Nuaimi et al., 2017).

Correction # 2

3. Method

3.3. Data collection procedure and instruments

Academic Integrity (AI) violation survey (Cronan et al., 2018), ICT literacy scale originally developed by Lau and Yuen (2014a) and extended as information ethics questionnaire by Nasser AL-Nuaimi et al. (2017) were benefitted in the current study as data gathering tools after modifying some items in relation to the research context.

Perceived behavioral control was assessed by a three-factor ICT literacy scale, in which the factors were conceptualized as Internet Literacy (INTL), Computer Literacy (COMPL), and Information Literacy (INFL). The scale was originally developed for middle school students by Lau and Yuen (2014a). Nasser AL-Nuaimi et al. (2017) extended, and adapted the scale by modifying items in a manner that conforms to the undergraduate context, and from which they obtained adequate reliability and validity values.

Correction # 3

5. Discussion

A contribution of the current study was that perceived behavioral control was assessed using three literacy measures. Similarly, Nasser AL-Nuaimi et al. (2017) substituted PBC with perceived ICT-literacy self-efficacy, though as a unitary construct, and have found a significant negative association between ICT-literacy self-efficacy and ICT-ethics-related behavior. Nevertheless, the results demonstrated that Internet literacy and computer literacy were not significant predictors.

Chan, H. F. (2015). Understanding adolescents' unethical online behaviors: A structural equation approach [Master's thesis, The University of Hong Kong]. http://hdl.handle.net/10722/208167

Nasser AL-Nuaimi, M., Bouazza, A., Abu-Hilal, M. M., & Al-Aufi, A. (2017). The psychometric properties of an information-ethics questionnaire. Performance Measurement and Metrics, 18(3), 166–179. https://doi.org/10.1108/PMM-10-2016-0044.

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